# serviceOrchestrationPushManagement

## Service Description

Service ID: *"serviceOrchestrationPushManagement"*

**Abstract**
This document describes the `serviceOrchestrationPushManagement` service, version 1.0.0, produced by the DynamicServiceOrchestration core system of Arrowhead (documentation edition 5.3.0).

# Contents

**1 Documentation edition and implementation scope** **3**

**2 Overview** **3**
2.1 Significant Prior Art . . . 3
2.2 How This Service Is Meant to Be Used . . . 3
2.3 Important Delimitations and Dependencies . . . 3

**3 Service Interface** **4**
3.1 operation subscribe . . . 4
3.2 operation unsubscribe . . . 4
3.3 operation trigger . . . 4
3.4 operation query . . . 4

**4 Information Model** **5**
4.1 struct MetadataRequirement . . . 5
4.2 struct OrchestrationJob . . . 5
4.3 struct OrchestrationNotifyInterface . . . 5
4.4 struct OrchestrationPushJobListResponse . . . 5
4.5 struct OrchestrationPushTrigger . . . 5
4.6 struct OrchestrationRequest . . . 6
4.7 struct OrchestrationServiceRequirement . . . 6
4.8 struct OrchestrationSubscriptionListRequest . . . 6
4.9 struct OrchestrationSubscriptionListResponse . . . 6
4.10 struct OrchestrationSubscriptionQueryRequest . . . 7
4.11 struct OrchestrationSubscriptionRequest . . . 7
4.12 struct OrchestrationSubscriptionResponse . . . 7
4.13 struct Page . . . 7
4.14 struct QoSRequirement . . . 8
4.15 Primitives . . . 8

**5 References** **9**

**6 Revision History** **10**
6.1 Amendments . . . 10
6.2 Quality Assurance . . . 10

# 1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

# 2 Overview

This document describes the `serviceOrchestrationPushManagement` service, which provides operator management of push orchestration: subscriptions made on behalf of consumers, and triggering of push orchestration. It is produced by the DynamicServiceOrchestration system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2]. The MQTT binding is specified in the MQTT/TLS/JSON IDD [3].

## 2.1 Significant Prior Art

Corresponds to the Arrowhead 4.x orchestration store push features.

## 2.2 How This Service Is Meant to Be Used

Operators and support systems (e.g. the Choreographer, PlantDescriptionEngine) create subscriptions for target systems, query and remove them, and trigger the execution of subscriptions; the orchestrator then sends the results to the subscribers' notify interfaces.

## 2.3 Important Delimitations and Dependencies

Push notifications are sent by worker threads (`push.orchestration.max.thread`); delivery is best effort and its outcome is recorded in the orchestration history.

# 3 Service Interface

This section describes the interfaces to the `serviceOrchestrationPushManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.

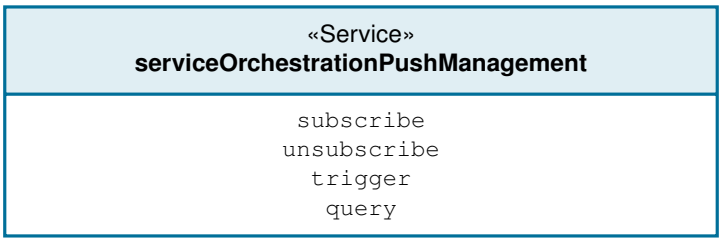



Figure 1: SysML-style block of the `serviceOrchestrationPushManagement` service and its operations.

The following interface operations are available.

## 3.1 operation subscribe (OrchestrationSubscriptionListRequest) : OrchestrationSubscriptionListResponse

Creates subscriptions on behalf of target systems.

## 3.2 operation unsubscribe (List<Identifier>)

Removes subscriptions by identifier.

## 3.3 operation trigger (OrchestrationPushTrigger) : OrchestrationPushJobListResponse

Starts push orchestration for the selected subscriptions (by target system or identifier); returns the started jobs.

## 3.4 operation query (OrchestrationSubscriptionQueryRequest) : OrchestrationSubscriptionListResponse

Returns a page of subscriptions matching the filter.

# 4 Information Model

Here, all data objects that can be part of the `serviceOrchestrationPushManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.15, which are used to represent things like hashes and identifiers.

## 4.1 struct MetadataRequirement

A JSON object of metadata requirements: each key is a metadata path (dot notation), each value the required value or an operator object such as `{"op": "LESS_THAN", "value": 10}`.

## 4.2 struct OrchestrationJob

One orchestration job (pull or push).

| Field | Type | Description |
|---|---|---|
| id | Identifier | Identifier (UUID) of the orchestration job. |
| status | String | Status of the job: `PENDING`, `IN_PROGRESS`, `DONE` or `ERROR`. |
| type | String | Orchestration type: `PULL` or `PUSH`. |
| requesterSystem | Name | Name of the system that requested the orchestration. |
| targetSystem | Name | Name of the consumer system the orchestration was made for. |
| serviceDefinition | Name | Service definition name. |
| subscriptionId | Identifier | Subscription that started the push job. |
| message | String | Status message, e.g. `1 local result: DataManager`. |
| createdAt | DateTime | UTC time of creation. |
| startedAt | DateTime | UTC time when the job started. |
| finishedAt | DateTime | UTC time when the job finished. |

## 4.3 struct OrchestrationNotifyInterface

Notification endpoint of push orchestration.

| Field | Type | Description |
|---|---|---|
| protocol | String | Protocol of the notification: `HTTP`, `HTTPS`, `MQTT` or `MQTTS`. |
| properties | Map<String, String> | For HTTP(S) `address`, `port`, `method`, `path`; for MQTT(S) `topic` (and broker `address`, `port`). |

## 4.4 struct OrchestrationPushJobListResponse

Push jobs started by a trigger.

| Field | Type | Description |
|---|---|---|
| jobs | List<OrchestrationJob> | Orchestration jobs started by the trigger. |

## 4.5 struct OrchestrationPushTrigger

Selects the subscriptions to execute now.

| Field | Type | Description |
|---|---|---|
| targetSystems | List<Name> | Consumers whose subscriptions are executed. |
| subscriptionIds | List<Identifier> | Subscriptions to execute. |

## 4.6 struct OrchestrationRequest

An orchestration request.

| Field | Type | Description |
|---|---|---|
| serviceRequirement | OrchestrationServiceRequirement | Requirements on the service the consumer wants to use. |
| orchestrationFlags | Map<String, Boolean> | Orchestration flags, e.g. `MATCHMAKING`, `ONLY_PREFERRED`, `ALLOW_INTERCLOUD`. |
| qualityRequirements | List<QoSRequirement> | Quality-of-service requirements (evaluated by DeviceQoSEvaluator). |
| exclusivityDuration | Number | Requested exclusive reservation of the provider in seconds. |

## 4.7 struct OrchestrationServiceRequirement

Requirements on the requested service.

| Field | Type | Description |
|---|---|---|
| serviceDefinition | Name | Service definition name. |
| operations | List<String> | Service operations the consumer wants to use. |
| versions | List<Version> | Accepted service versions. |
| alivesAt | DateTime | Only instances that are still registered (not expired) at this UTC time are matched. |
| metadataRequirements | List<MetadataRequirement> | Requirements on the service instance metadata; each map is one alternative. |
| interfaceTemplateNames | List<String> | Accepted interface templates, e.g. `generic_https`. |
| interfaceAddressTypes | List<String> | Address types the provider interface must offer. |
| interfaceProperty Requirements | List<MetadataRequirement> | Requirements on the interface properties; each map is one alternative. |
| securityPolicies | List<String> | Accepted interface security policies. |
| preferredProviders | List<String> | Provider systems to prefer. |

## 4.8 struct OrchestrationSubscriptionListRequest

Subscriptions to create on behalf of consumers.

| Field | Type | Description |
|---|---|---|
| subscriptions | List<OrchestrationSubscription Request> | Subscriptions to create. |

## 4.9 struct OrchestrationSubscriptionListResponse

List of subscriptions.

| Field | Type | Description |
|---|---|---|
| entries | List<OrchestrationSubscription Response> | Matching records of the current page. |
| count | Number | Total number of matching records (all pages). |

## 4.10 struct OrchestrationSubscriptionQueryRequest

Paged filter for subscriptions.

| Field | Type | Description |
|---|---|---|
| pagination | Page | Paging and sorting of the result. |
| ownerSystems | List<String> | Subscription owner names to match. |
| targetSystems | List<Name> | Target system names. |
| serviceDefinitions | List<Name> | Service definition names to match. |

## 4.11 struct OrchestrationSubscriptionRequest

A push orchestration subscription.

| Field | Type | Description |
|---|---|---|
| targetSystemName | Name | Consumer system; default the requester. |
| orchestrationRequest | OrchestrationRequest | The orchestration request executed for the subscription. |
| notifyInterface | OrchestrationNotifyInterface | Where and how the orchestration result is pushed to the target system. |
| duration | Number | Requested lifetime of the subscription in seconds; no value means unlimited. |

## 4.12 struct OrchestrationSubscriptionResponse

A stored subscription.

| Field | Type | Description |
|---|---|---|
| id | Identifier | Identifier (UUID) of the subscription. |
| ownerSystemName | Name | Name of the system that created the subscription. |
| targetSystemName | Name | Name of the consumer system the push orchestration is made for. |
| orchestrationRequest | OrchestrationRequest | The orchestration request executed for the subscription. |
| notifyInterface | OrchestrationNotifyInterface | Where and how the orchestration result is pushed to the target system. |
| expiredAt | DateTime | UTC time when the subscription expires. |
| createdAt | DateTime | UTC time of creation. |

## 4.13 struct Page

Paging and sorting; all fields optional.

| Field | Type | Description |
|---|---|---|
| page | Number | Zero-based page number. |
| size | Number | Number of records per page (at most `max.page.size`, default 1000). |
| direction | String | Sort direction, `ASC` or `DESC`. |
| sortField | String | Field to sort by. |

## 4.14 struct QoSRequirement

One quality-of-service requirement.

| Field | Type | Description |
|---|---|---|
| type | String | Type of the QoS evaluation, as known by DeviceQoSEvaluator. |
| operation | String | QoS operation: `FILTER` or `SORT`. |
| requirements | Map<String, Object> | Parameters of the QoS requirement. |

## 4.15 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

| Type | Description |
|---|---|
| Boolean | One out of `true` or `false`. |
| DateTime | Pinpoints a specific moment in time (UTC). |
| Identifier | A string that identifies a record, e.g. a UUID or a composite identifier. |
| List | An *array* of a known number of items, each having type A. |
| Map | A set of key-value pairs. |
| Name | A string identifier that is intended to be both human and machine-readable. |
| Number | An integer number. |
| Object | Any structured value. |
| String | A chain of characters. |
| Version | Specifies a version, semantic versioning `major.minor.patch`. |

# 5 References

[1] J. Delsing, “DynamicServiceOrchestration – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, serviceorchestration-dynamic/documentation/v5.3.0/DynamicServiceOrchestration-v5.3.0-SysD.pdf.

[2] ——, “serviceOrchestrationPushManagement HTTP/TLS/JSON – Interface Design Description (DynamicServiceOrchestration),” Eclipse Arrowhead, IDD 5.3.0, 2026, dynamicServiceOrchestration-v5.3.0-serviceOrchestrationPushManagement-HTTP-IDD.pdf.

[3] ——, “serviceOrchestrationPushManagement MQTT/TLS/JSON – Interface Design Description (DynamicServiceOrchestration),” Eclipse Arrowhead, IDD 5.3.0, 2026, dynamicServiceOrchestration-v5.3.0-serviceOrchestrationPushManagement-MQTT-IDD.pdf.

# 6 Revision History

## 6.1 Amendments

| No. | Date | Version | Subject of Amendments | Author |
|---|---|---|---|---|
| 1 | 2026-10-22 | 5.2.2 | Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace. | Jerker Delsing |
| 2 | 2026-11-09 | 5.3.0 | New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained. | Documentation maintenance |

## 6.2 Quality Assurance

| No. | Date | Version | Approved by |
|---|---|---|---|
| 1 | | 5.3.0 | |